Multiple-Impulse Spacecraft Rendezvous Optimization

Report Outline

Hasnain

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Abstract

Writing prompt: Write this last. In one paragraph, state the rendezvous problem, the model and search method used, the main quantitative result, and the most important conclusion or limitation. Leave results unspecified until you have verified them.

1 Introduction and Aim

1.1 The rendezvous problem

Writing prompt: Explain the roles of the chaser and target, and why meeting requires matching position and velocity at the same time. Briefly motivate the use of a small number of engine burns.

1.2 Project aim and research question

A proposed aim is to develop and evaluate a simplified two-dimensional simulation of spacecraft rendezvous using multiple impulsive burns, and to investigate how burn choices influence final position error, velocity error, and total velocity change.

Central question: How effectively can a small number of burns achieve rendezvous within the chosen model and search limits?

Writing prompt: State the specific scenario investigated and what your report will demonstrate. If you compare accuracy and total velocity change, identify that trade-off here. Describe the work completed rather than promising experiments that have not been done.

2 Physical Model and Problem Definition

2.1 Assumptions and equations of motion

- Planar motion in an Earth-centred inertial Cartesian frame.
- Central gravity with constant gravitational parameter μ.
- An uncontrolled target and a chaser with instantaneous burns.
- State any other assumptions and omitted physical effects.

For each spacecraft, use position $\mathbf{r} = (x, y)$ and velocity $\mathbf{v} = (v_x, v_y)$:

$$\dot{\mathbf{r}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = -\mu \frac{\mathbf{r}}{\|\mathbf{r}\|^3}. \tag{1}$$

At burn time t_i, position is continuous and velocity changes according to

$$\mathbf{v}(t_i^+) = \mathbf{v}(t_i^-) + \Delta\mathbf{v}_i. \tag{2}$$

Writing prompt: Explain these equations physically. The current rendezvous scripts restrict burns to the instantaneous velocity direction, using a signed magnitude for prograde or retrograde burns. Describe this restriction if you use those scripts; do not imply that arbitrary directions were searched.

2.2 Scenario and definition of success

Writing prompt: Specify both initial states, units, final time T, number of burns, and allowed burn magnitudes and times. Define acceptable final separation and relative speed before interpreting a result as successful rendezvous.

$$e_r = \|\mathbf{r}_c(T) - \mathbf{r}_t(T)\|, \tag{3}$$

$$e_v = \|\mathbf{v}_c(T) - \mathbf{v}_t(T)\|, \tag{4}$$

$$\Delta v_{\text{total}} = \sum_{i=1}^{N} \|\Delta \mathbf{v}_i\|. \tag{5}$$

Writing prompt: Distinguish final separation from the minimum separation during the trajectory. Describe total Δv as a measure of manoeuvre effort; it is not a direct measurement of propellant mass.

3 Computational Method

3.1 Trajectory propagation

Writing prompt: Describe the numerical solver, tolerances, target propagation, and how chaser propagation stops and restarts at each burn. Explain how the states of both spacecraft are compared at the same final time.

3.2 Objective function and search

The intended weighted objective has the form

$$J = w_r e_r^2 + w_v e_v^2 + w_{\Delta v} \Delta v_{\text{total}}. \tag{6}$$

Writing prompt: Check that the implemented objective matches this expression, particularly how burn magnitudes are extracted from stored burn/time pairs. Report weights and their scaling: squared position error, squared velocity error, and total Δv have different units. A low weighted cost alone does not establish that both rendezvous tolerances were met.

- Identify the decision variables and their bounds.
- Explain differential evolution at a practical level.
- State how burn ordering and invalid candidates are handled.
- Record the random seed, search budget, and stopping criteria.
- Include GPU acceleration only if it forms part of the evaluated work; support any speed claim with comparable measured runs.

4 Simulation Checks

Writing prompt: Present checks actually performed and their outcomes. Useful checks include a circular-orbit reference case, conservation of energy and angular momentum between burns, and repeating a candidate trajectory with tighter solver tolerances. Explain what each check establishes.

- Suggested figure: reference orbit or conservation error over time.
- Report whether numerical differences are small relative to the rendezvous tolerances used in the report.

5 Experiments and Results

5.1 Baseline and best trajectory found

Writing prompt: Compare a clearly defined baseline, such as no burns or a manual strategy, with the best sequence found for the same initial conditions and final time. Fill the table with verified values and identify units.

Table 1: Rendezvous performance. Blank cells await verified results.

Strategy	e_r (m)	e_v (m/s)	Total Δv (m/s)
Baseline			
Best sequence found			

- Plot both trajectories with burn locations and final positions.
- Plot separation and relative speed against time, marking burns.
- List each selected burn's time, signed magnitude, and direction.
- State explicitly whether both success tolerances were met.

5.2 Selected comparisons

Writing prompt: Choose only comparisons that answer your research question and fit the available time. Possible variables include burn count, objective weights, or search budget. Keep other conditions fixed. For a stochastic search, use repeated seeds if assessing reliability. Report unsuccessful cases when they help explain a limitation.

6 Discussion

Writing prompt: Explain what the results mean, connecting trajectory changes to the physics rather than simply describing plots.

- Why did the chosen burn sequence improve the rendezvous attempt?
- How did timing and velocity changes affect relative motion?
- What trade-offs are supported by the experiments?
- Were remaining errors associated with numerical accuracy, search limits, burn restrictions, or objective weighting? Distinguish evidence from possible explanations.
- What do the simplified dynamics omit, and how does that limit conclusions about a real mission?

Writing prompt: Use "best solution found within the tested search" unless you have evidence of global optimality. Avoid general conclusions about burn count or fuel savings from a single case.

7 Conclusion and Future Work

Writing prompt: Answer the central question directly. Summarize the strongest quantitative finding, the main physical insight, and the principal limitation. Suggest one or two specific next steps justified by the results.

References

Writing prompt: Cite the project brief, any external theory used, numerical-method documentation, and any adapted code. Add bibliographic entries as you write.

A Reproducibility Details

Writing prompt: Record the script/version used, software dependencies, exact initial states, solver and optimizer settings, seeds, objective weights, and final burn sequence. Include supporting checks or additional plots here when they would interrupt the main argument.